Exc no: 4

03/08/26

# Setting Up a Local Area Network (LAN)

## Aim:

To design, set up and configure a local Area Network (LAN) connecting four computers through a network switch and demonstrate resource/data sharing across the network.

## Equipment required

| Equipment | Quantity | specification |
|---|---|---|
| Personal computer | 4 | Ethernet (NIC) enabled |
| Network Switch | 1 | 8/16/24 port |
| Ethernet Cables (Cat5e/Cat6) | 4 | straight through, RJ-45 connectors |

## IP address configuration table

| Device | IP Address | Subnet mask | Connected To |
|---|---|---|---|
| PC0 | 10.1.1.1 | 255.0.0.0 | Switch Port Fa0/1 |
| PC1 | 10.1.1.2 | 255.0.0.0 | Switch Port Fa0/2 |
| PC2 | 10.1.1.3 | 255.0.0.0 | Switch Port Fa0/3 |
| PC3 | 10.1.1.4 | 255.0.0.0 | Switch Port Fa0/4 |
| Switch | 10.1.1.5 | 255.0.0.0 | VLAN 1 Interface |

## Implementation in CISCO Packet Tracer:

1) Place the Devices
   - Open cisco packet tracer
   - From the device category panel, select End Devices and drag 4 PCs onto the workspace.
   - Select switches and drag 1 switch.

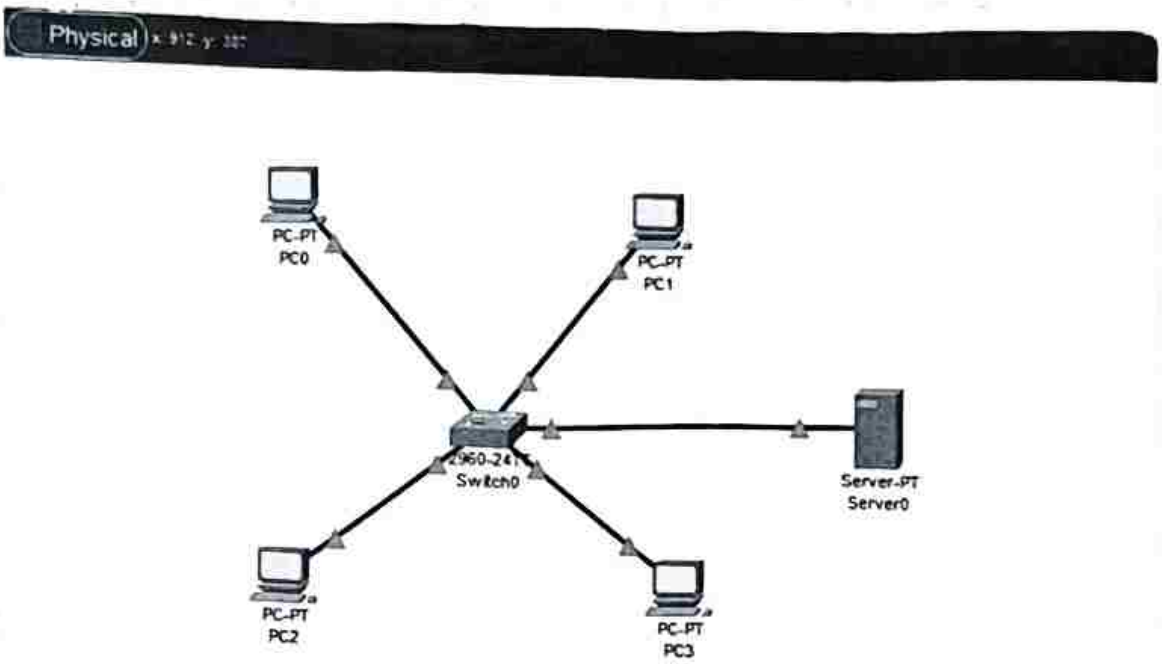
2) Cable the Devices
- Click Connections → select Copper Straight - Through cable
- Click each PC, then click the switch, and select an available Fast Ethernet port to complete each connection.
- Repeat for all four PCs (ports Fa0/1 to Fa0/4). Green link lights confirm a successful physical connection.

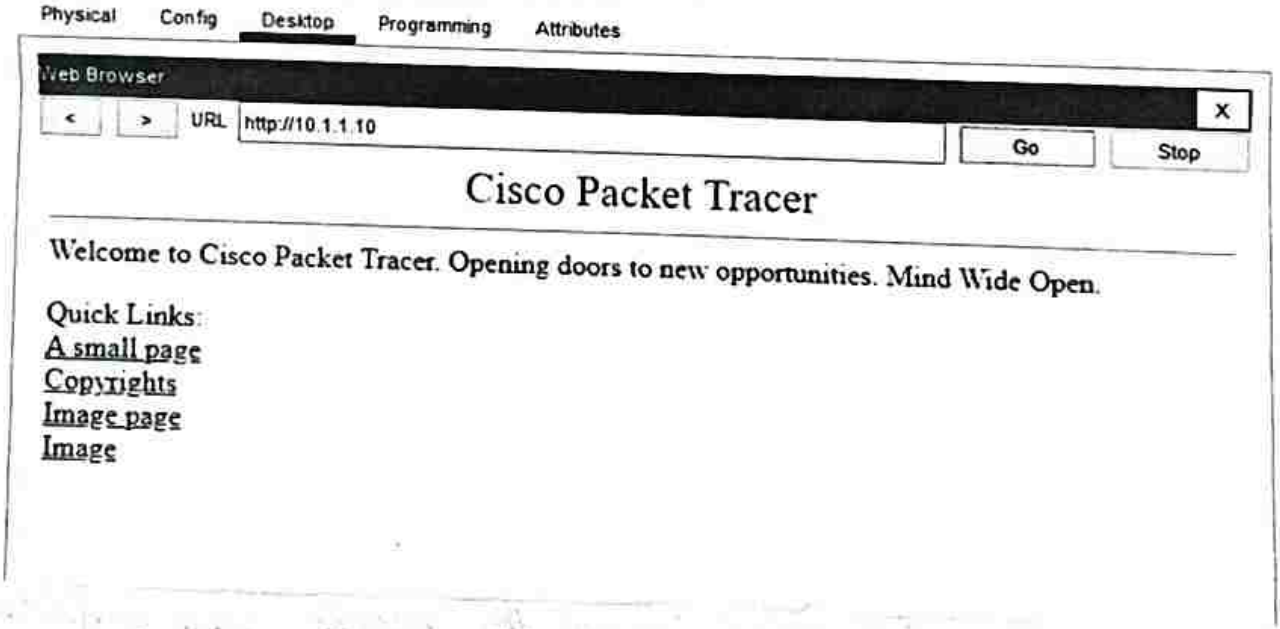
3) Assign IP Addresses to PCs
- Click a PC → Desktop tab - IP configuration
- Enter the Static IP address and Subnet Mask as per the IP table.

4) Configure the Switch Management IP
- Click the switch → CLI tab
- enable
  configure terminal
  interface vlan 1
  ip address 10.1.1.5 255.0.0.0
  no shutdown
  exit

5) Verify connectivity using Ping
- Click a PC → Desktop tab → Command Prompt
- Use the ping to test connectivity.
- Successful message.

6) Demonstrate Data Sharing (Server Simulation).
- Add a server device (End Devices → Server-PT) and connect it to the switch
- Assign it an IP address
- Open the server's Services tab & enable the HTTP
- From another PC, open Web browser and enter the server's IP address to load the page, confirming successful data transfer across the LAN.




Result:

Thus lan is setted up and connection is verified successfully.